Bayesian Stopping Rules for Machine Learning-Assisted Systematic Review Screening

A Bayesian Framework for Robust Stopping Criteria

Master Thesis

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Abstract

Machine learning-assisted prioritization has substantially reduced the workload of systematic review screening, but determining when it is safe to stop screening remains an open methodological problem. This thesis develops and evaluates the first Bayesian stopping rule for machine learning-assisted systematic review screening. The screening process is modeled as a biased urn process in which both the total number of relevant documents and the ranking bias parameter are treated as unknown quantities to be inferred jointly from the observed screening sequence. Stopping is triggered once the posterior probability of having achieved a recall target level exceeds a user-specified confidence threshold. The method is evaluated through Monte Carlo simulations across four experimental configurations and applied to real-world screening data from ten completed systematic reviews. Results show that the Bayesian method achieves consistently lower miss rates than a frequentist baseline at every confidence level tested, at the cost of substantially lower work savings. A robustness analysis under time-varying bias reveals that the method is sensitive to violations of its constant bias assumption in a structured and predictable way, and that a simple conservative adjustment to the likelihood extends the range of assumption violations under which the recall guarantee is maintained. To the best of the author's knowledge, this is the first stopping criterion for systematic review screening grounded in a fully Bayesian framework.

Contents

1 Introduction **1**

2 Literature Review **2**
2.1 Machine Learning-Assisted Screening in Evidence Synthesis 2
2.2 The Stopping Problem Under Uncertainty 3
2.3 Hypergeometric Foundations of Statistical Stopping 4
2.4 Biased Urn Models and Noncentral Hypergeometric Distributions 5
2.5 Bayesian Approaches in Evidence Synthesis 6
2.6 Gap in the Literature 6

3 Methodology **7**
3.1 Problem Setup 7
3.2 Generative Model: The Biased Urn Process 7
3.3 Bayesian Inference: Posterior Updating 8
3.4 Stopping Rule 9
3.5 Computational Implementation 9
3.6 Monte Carlo Simulation Study 10
3.7 Experimental Design 10

4 Results **12**
4.1 Controlled Simulation under Constant Bias 12
4.1.1 Recall at Stopping 12
4.1.2 Work Saved 13
4.1.3 Miss Rate and Guarantee Reliability 14
4.2 Simulation Study: Changing Bias 15
4.2.1 Sensitivity to Decay Strength 15
4.2.2 Conservative Adjustment 17
4.3 Performance on Real-World Screening Data 17
4.3.1 Reliability 17
4.3.2 Efficiency 18
4.3.3 Review-Level Heterogeneity 18

5 Discussion **20**
5.1 A Conservative but Trustworthy Stopping Criterion 20
5.2 The Reliability–Efficiency Trade-off Across Reviews 21
5.3 Robustness to Changing Bias and Conservative Adjustment . . . 22
5.4 First Bayesian Stopping Rule for Systematic Review Screening . . 23
5.5 Limitations and Future Work 24

6 Conclusion **25**

Appendix: Code Repository **28**

1 Introduction

Systematic reviews occupy a central role in evidence-based research by synthesizing findings from large bodies of scientific literature. However, the identification of relevant studies for inclusion in a systematic review typically requires screening thousands of titles and abstracts, making the process both time-consuming and resource-intensive (O'Mara-Eves et al., 2015). The problem is compounded by the continuous growth of the published literature: the volume of scientific output has expanded to the point where keeping pace with new evidence through manual review alone is increasingly impractical (Bastian et al., 2010). In response, machine learning-assisted screening methods have been proposed to accelerate this process by ranking documents according to their predicted relevance, allowing reviewers to prioritize screening documents that are more likely to be relevant (Cormack & Grossman, 2014).

While machine learning (ML) ranking can substantially reduce the workload associated with systematic review screening, an important methodological question remains: when is it safe to stop screening? Stopping prematurely can leave a significant share of the relevant literature unidentified, with downstream consequences for the validity of the synthesis (M. Callaghan et al., 2024), while screening all documents removes the efficiency gains offered by ML methods. The absence of a trusted stopping criterion has concrete consequences for practice: the Cochrane Handbook for Systematic Reviews of Interventions — the authoritative methodological guide for evidence synthesis — does not currently recommend any stopping criterion for ML-assisted screening, reflecting ongoing concerns that existing methods cannot reliably guarantee that a sufficient proportion of relevant studies has been identified (M. Callaghan et al., 2024). This problem is not unique to systematic reviews. In legal discovery, technology-assisted review must identify all disclosable documents; in hate speech moderation and fraud detection, ranked document collections must be screened until reviewers can be confident that substantially all relevant instances have been found. In each of these contexts, the need for a stopping criterion that provides a genuine statistical guarantee is both practically important and methodologically unresolved. To address this challenge, several approaches have been proposed that aim to estimate the proportion of relevant documents already identified during screening, commonly referred to as recall. In practice, systematic review guidelines typically require a high recall threshold (for example, 95%) to ensure that the majority of relevant studies have been identified (Higgins et al., 2019).

A promising approach to this problem is the use of Bayesian stopping rules, which provide a probabilistic guarantee that recall has exceeded a predefined target threshold. In this framework, the screening process is modeled as a sequential sampling problem in which documents are screened one at a time. As screening progresses, Bayesian inference is used to update the posterior distribution over unknown quantities: the total number of relevant documents and the bias intro-

duced by the ranking model (Gelman et al., 2013). Screening can then be stopped once the posterior probability that recall exceeds a target threshold becomes sufficiently large (M. W. Callaghan & Müller-Hansen, 2020).

In this thesis, a Bayesian stopping criterion for ML-assisted systematic review screening is implemented and evaluated using simulation experiments. The screening process is modeled as a biased urn process, in which relevant documents are more likely to appear earlier in the screening sequence due to the ranking produced by a ML classifier (M. W. Callaghan & Müller-Hansen, 2024). The degree of ranking effectiveness is captured by a bias parameter that determines how strongly relevant documents are prioritized during screening. A Bayesian inference procedure is used to sequentially update the posterior distribution over the total number of relevant documents and the bias parameter based on the observed screening outcomes (Gelman et al., 2013).

To evaluate the performance of the proposed stopping rule, a series of Monte Carlo simulation experiments are conducted. These simulations generate synthetic screening sequences under controlled conditions and measure the resulting recall at the stopping point as well as the proportion of screening effort saved (Repke et al., 2026). Particular attention is given to the lower tail of the recall distribution to assess whether the stopping rule successfully satisfies the desired probabilistic recall guarantee.

This thesis develops, evaluates, and tests a Bayesian stopping rule for ML-assisted systematic review screening. A computational implementation using vectorized algorithms is developed to enable efficient large-scale simulation experiments. The performance is then evaluated through Monte Carlo simulations across a range of experimental settings, before examining its robustness under scenarios where the effectiveness of the ranking model changes over the course of the screening process. Finally, the method is applied to real-world screening data from a collection of completed systematic reviews enabling a systematic comparison of the Bayesian stopping rule against a frequentist baseline across multiple reviews and confidence thresholds, evaluated on a high-performance computing cluster to enable analysis at scale. Together, these contributions provide both a theoretical validation and an empirical assessment of Bayesian stopping in realistic review conditions.

2 Literature Review

2.1 Machine Learning-Assisted Screening in Evidence Synthesis

The rapid growth of scientific literature has placed substantial strain on systematic review processes. Screening titles and abstracts is among the most labor-intensive phases of evidence synthesis, consuming substantial reviewer time and

effort (Shemilt et al., 2013).

ML-assisted prioritization has emerged as a promising solution. Active learning approaches rank documents by predicted relevance, allowing reviewers to identify relevant studies earlier in the screening process (O'Mara-Eves et al., 2015). Numerous empirical studies demonstrate that prioritized screening can achieve high recall while substantially reducing workload (Yang et al., 2022).

However, while ranking methods have improved, a critical methodological bottleneck remains: determining when it is safe to stop screening. This stopping problem is the central unsolved challenge in practical application (M. Callaghan et al., 2024). The absence of a trusted stopping criterion limits the practical adoption of ML-assisted screening, a gap this thesis directly addresses.

The potential for workload reduction through ML-assisted prioritization has been demonstrated across multiple domains. In their systematic review of text mining approaches, O'Mara-Eves et al. (2015) report that screening workload reductions in the 30–70% range are potentially achievable, though the magnitude varies substantially with prevalence, corpus size, and the quality of the ranking model. This variability is itself an important finding: reviews with very effective ranking models realize the largest efficiency gains, while those with sparser evidence bases or weaker classifiers see more modest savings. Both the potential for efficiency gain and its dependence on ranking quality are directly relevant to the results presented in this thesis.

2.2 The Stopping Problem Under Uncertainty

In live review settings, recall is fundamentally unobservable until all documents have been screened (M. W. Callaghan & Müller-Hansen, 2020). Reviewers must therefore decide whether to stop while remaining uncertain about the total number of relevant studies. This creates a decision problem under uncertainty: stopping too early risks missing relevant evidence and introducing bias; stopping too late forfeits potential work savings. Crucially, the efficiency gains reported in existing studies may not translate to new reviews, as they often rely on retrospective data where the optimal stopping point is already known (M. Callaghan et al., 2024).

Several authors emphasize that existing systems lack robust stopping criteria, limiting their applicability in real-world review contexts. Simulation studies further confirm that screening performance varies substantially across review topics, prevalence levels, and model configurations, underscoring the difficulty of defining a universally safe stopping point (Ferdinands et al., 2023). In particular, heuristic rules lack formal statistical validity, prevalence estimators ignore sampling uncertainty, and model-dependent criteria remain unreliable under imperfect ranking (M. W. Callaghan & Müller-Hansen, 2020; Repke et al., 2026).

Alternative approaches have sought to address these limitations through statistical modeling of the remaining relevant document population. Bron et al. (2025) propose using Chao's estimator — a population size estimator with ori-

gins in capture-recapture ecology — as a stopping criterion. On the other hand, Stevenson and Bin-Hezam (2023) model the occurrence of relevant documents as a point process. Despite this diversity of approaches, no consensus has emerged on a universally reliable method (Repke et al., 2026).

A recent large-scale evaluation by Repke et al. (2026) reports that the majority of existing stopping rules either fall short of the intended recall guarantee or fail to deliver meaningful work savings. Heuristics based on consecutive exclusions perform particularly poorly, with most relevant records left unidentified (Repke et al., 2026), and M. Callaghan et al. (2024) argue forcefully that ad hoc stopping rules are incompatible with the methodological standards expected of high quality systematic reviews.

A recurring theme in the literature is that reliable stopping rules must: (1) explicitly quantify uncertainty; (2) allow specification of a recall target level; (3) provide interpretable confidence statements; and (4) remain robust to imperfect ranking performance (M. Callaghan et al., 2024; M. W. Callaghan & Müller-Hansen, 2020). Despite nearly two decades of research in active learning for systematic reviews, early stopping remains controversial and is not universally recommended in review guidelines (M. W. Callaghan & Müller-Hansen, 2024).

2.3 Hypergeometric Foundations of Statistical Stopping

A major methodological advance was the formulation of stopping as a hypothesis testing problem grounded in the hypergeometric distribution. When screening without replacement from a finite corpus of size N, the number of relevant documents observed after screening n records follows a hypergeometric distribution under random sampling. This finite-population model provides a principled basis for reasoning about the number of remaining relevant documents, and grounds stopping rules in finite-population inference rather than asymptotic binomial reasoning (M. W. Callaghan & Müller-Hansen, 2020).

Stopping can be formulated as a hypothesis test where the null hypothesis asserts that the achieved recall is below a target threshold — equivalently, that the number of remaining relevant documents exceeds a critical value (M. W. Callaghan & Müller-Hansen, 2020). Formally:

$$H_0 : \texttt{Recall} < \tau \tag{1}$$

where τ is a predefined recall target. Such methods allow reviewers to reject the null hypothesis with controlled Type I error, providing a formal statistical guarantee (M. W. Callaghan & Müller-Hansen, 2020).

However, prioritized screening violates the central hypergeometric assumption of random sampling. Since documents are ranked by predicted relevance, early draws are systematically biased toward relevant studies (M. W. Callaghan & Müller-Hansen, 2024). Applying a central hypergeometric model in this context therefore leads to conservative stopping behavior: the test implicitly assumes

a worst-case prior over remaining relevant documents, requiring more evidence before rejecting H_0 than is strictly necessary (M. W. Callaghan & Müller-Hansen, 2024). This over-conservatism has been empirically confirmed (Repke et al., 2026).

2.4 Biased Urn Models and Noncentral Hypergeometric Distributions

To address this mismatch, recent work has introduced biased urn formulations. A succession of irrelevant documents is an intuitive signal that few relevant records remain unscreened, but their probative weight for recall is shaped by both the count of relevant records already retrieved and the size of the unscreened residual (M. W. Callaghan & Müller-Hansen, 2024). Biased urn models make this reasoning precise.

Noncentral hypergeometric distributions generalize the central hypergeometric to account for differential selection probabilities (Fog, 2008a):

- *Wallenius' noncentral hypergeometric distribution* models sequential biased sampling without replacement, where selection probabilities update as items are drawn.
- *Fisher's noncentral hypergeometric distribution* arises from conditioning independent biased draws on a fixed total sample size.

Both reduce to the standard hypergeometric when the odds ratio equals one (i.e., no bias between classes) (Fog, 2008a, 2008b).

Empirical evaluation by Repke et al. (2026) identifies the hypergeometric criterion of M. W. Callaghan and Müller-Hansen (2020) as the only method in their benchmark that reliably meets the specified recall target across simulations. However, the same evaluation finds it to be markedly conservative, particularly at high recall targets and in low-prevalence settings, leaving substantial efficiency gains unrealized. This conservatism is a direct consequence of the random-sampling assumption embedded in the underlying hypergeometric model. BUSCAR (Biased Urns for Stopping Criteria in Assisted Reviews) addresses this limitation by extending hypergeometric stopping with an urn framework that explicitly models the bias introduced by prioritized screening, rather than assuming random draws (M. W. Callaghan & Müller-Hansen, 2024).

A further limitation of the biased urn approach concerns the estimation of the bias parameter ω itself. In practice, ω is not directly observable and must be specified by the user prior to screening. Existing implementations have relied on rough heuristics or conservative default values, effectively amounting to an educated guess (M. W. Callaghan & Müller-Hansen, 2024). This is problematic because the stopping behavior of the rule is sensitive to the assumed value of ω: an overestimated bias leads to overly optimistic recall estimates and premature stopping, while an underestimated bias produces unnecessary conservatism.

A Bayesian framework naturally addresses this limitation by treating ω as an unknown parameter to be inferred jointly with K from the observed screening sequence, rather than fixed in advance.

More broadly, across the literature existing approaches lack calibrated uncertainty quantification and frame stopping as a frequentist hypothesis test rather than a posterior inference problem, meaning reviewers cannot directly specify a confidence level for their recall guarantee (M. W. Callaghan & Müller-Hansen, 2020). Researchers have explicitly called for uncertainty-aware methods capable of estimating compounding effects through both ranking imperfection and sampling variability (Repke et al., 2026), a gap that the Bayesian framework proposed in this thesis directly addresses.

2.5 Bayesian Approaches in Evidence Synthesis

Bayesian methods have been implemented in related areas of evidence synthesis, including meta-analysis (Gelman et al., 2013) and the modeling of publication bias, but their use in the stopping problem for prioritized screening remains largely unexplored. For instance, Bayesian hierarchical models are routinely used in meta-analysis to pool effect estimates across studies while accounting for between-study heterogeneity (Gelman et al., 2013), and Bayesian approaches have been proposed for the detection of publication bias in evidence synthesis pipelines. The closest existing work by M. W. Callaghan and Müller-Hansen (2020) introduces a confidence-based stopping framework using frequentist hypothesis testing, but stop short of a fully Bayesian generative model of the screening process itself. Rather than constructing a hypothesis test about recall under a worst-case assumption about remaining relevant documents, the framework in this thesis proposes to treat the unknown quantities, which are the total number of relevant documents K and the ranking bias ω, as parameters with prior distributions that are updated as each screening outcome is observed (Gelman et al., 2013). This produces a posterior distribution over recall at any given point in screening that supports direct probabilistic statements, selected confidence thresholds, and explicit quantification of uncertainty. These are features that frequentist stopping rules can only approximate indirectly (Berger, 1985).

2.6 Gap in the Literature

Although hypergeometric and biased urn models have been developed within a frequentist framework, a fully Bayesian stopping rule for prioritized screening that jointly infers both the total number of relevant documents and the ranking bias from observed data remains underexplored. The field has yet to converge on a stopping rule that is simultaneously safe, robust, and able to deliver meaningful work savings at user-specified confidence levels (Repke et al., 2026). Specifically, the literature lacks posterior modeling of the total number of relevant

documents under biased sampling; explicit credible statements about recall with user-specified confidence levels; and systematic empirical evaluation of posterior-based stopping across multiple review scenarios and prevalence settings. This thesis presents the first stopping method for systematic review screening to adopt a Bayesian framework, addressing the lack of calibrated uncertainty quantification and user-specified confidence levels that has been repeatedly identified as a shortcoming of existing approaches (M. Callaghan et al., 2024).

3 Methodology

3.1 Problem Setup

In systematic review screening, documents are ranked by a ML model that estimates the probability of relevance, and reviewers screen documents sequentially in descending order of predicted relevance (Cormack & Grossman, 2014). The objective is to develop a stopping rule that allows screening to terminate early while guaranteeing, with high probability, that recall exceeds a predefined target threshold.

Let N denote the total number of documents in the corpus and K the unknown total number of relevant documents. After screening n documents, let k_n denote the number of relevant documents identified. Recall at step n is then defined as:

$$R_n = \frac{k_n}{K} \tag{2}$$

Since K is unknown in a live review, R_n is unobservable in practice (M. W. Callaghan & Müller-Hansen, 2020). Stopping therefore requires reasoning under uncertainty about K given the data observed so far.

3.2 Generative Model: The Biased Urn Process

The screening process is modeled as a biased urn process, in which documents are drawn sequentially without replacement from a corpus of N documents, of which K are relevant (Johnson & Kotz, 1977). Under unbiased (random) sampling, the number of relevant documents encountered after screening n records would follow a standard hypergeometric distribution (M. W. Callaghan & Müller-Hansen, 2020). However, prioritized screening systematically draws relevant documents earlier, violating this assumption (M. W. Callaghan & Müller-Hansen, 2024).

To capture this bias, a Wallenius' noncentral hypergeometric distribution is adopted as the sampling model (Fog, 2008b). At each step, the probability of drawing a relevant document is:

$$p_{\text{rel}} = \frac{\omega \cdot R_{\text{remaining}}}{\omega \cdot R_{\text{remaining}} + I_{\text{remaining}}} \tag{3}$$

where $R_{\text{remaining}}$ and $I_{\text{remaining}}$ denote the number of remaining relevant and irrelevant documents respectively, and $\omega > 0$ is a bias parameter capturing the strength of the ranking model. When $\omega = 1$ the process reduces to random sampling; when $\omega > 1$ relevant documents are drawn earlier, consistent with effective ML-assisted prioritization. The parameter ω thus serves as a proxy for ranking quality, and its uncertainty is explicitly modeled alongside K.

3.3 Bayesian Inference: Posterior Updating

Rather than testing a null hypothesis about recall under worst-case assumptions, the stopping problem is reformulated here as a posterior inference problem. Specifically, the approach:

1. Places a prior over the unknown total number of relevant documents K and the bias parameter ω
2. Updates this prior sequentially using a likelihood that accounts for biased sampling
3. Computes the posterior probability of having achieved a recall target level
4. Allows the user to directly specify a confidence threshold γ at which screening is stopped.

This reframing replaces fixed significance tests with calibrated probabilistic statements about recall (Berger, 1985; Gelman et al., 2013).

The central inferential task is to characterize the joint posterior distribution over the unknown parameters K and ω given the observed screening sequence $y_{1:n}$, where $y_i \in \{0, 1\}$ indicates whether the i-th document screened was irrelevant or relevant.

Following the principles of Bayesian updating (Gelman et al., 2013), the posterior after n steps is:

$$p(K, \omega \mid y_{1:n}) \propto p(y_n \mid K, \omega, y_{1:n-1})\, p(K, \omega \mid y_{1:n-1}) \tag{4}$$

As screening progresses, the posterior is updated sequentially: each new observation shifts probability mass toward values of K and ω consistent with the observed pattern of relevant and irrelevant documents (Gelman et al., 2013). This replaces the implicit worst-case assumptions of frequentist hypergeometric tests with an explicit, evolving belief distribution (M. W. Callaghan & Müller-Hansen, 2024).

A uniform (non-informative) prior is placed over a discrete grid of values for $K \in \{k_n, k_n + 1, \ldots, N\}$ and $\omega \in \{\omega_{\min}, \ldots, \omega_{\max}\}$, consistent with standard practice for grid-based Bayesian inference (Gelman et al., 2013). The likelihood at each step is evaluated using Wallenius' noncentral hypergeometric probability

mass function (Fog, 2008a). Crucially, by placing a prior over ω and updating it alongside K, the method infers the ranking bias directly from the observed screening sequence, rather than requiring the user to specify it in advance - directly addressing the limitation identified in existing biased urn approaches (M. W. Callaghan & Müller-Hansen, 2024).

3.4 Stopping Rule

Let $\tau \in (0, 1]$ be a recall target level and $\gamma \in (0, 1]$ a desired confidence level. The Bayesian stopping rule stops the screening in the first step n such that:

$$\Pr(R_n \geq \tau \mid y_{1:n}) \geq \gamma \tag{5}$$

This posterior probability is computed by marginalizing over the joint posterior of K and ω:

$$\Pr(R_n \geq \tau \mid y_{1:n}) = \sum_{K,\omega} \mathbf{1}\left[\frac{k_n}{K} \geq \tau\right] p(K, \omega \mid y_{1:n}) \tag{6}$$

Stopping is triggered when this quantity first exceeds γ. Unlike frequentist stopping rules, which test a null hypothesis at a fixed significance level, this formulation directly answers the question of practical interest: with what probability has a given recall target already been achieved? (Berger, 1985). Both τ and γ are user-specified, enabling reviewers to calibrate the trade-off between recall assurance and workload reduction (M. W. Callaghan & Müller-Hansen, 2020). Moreover, Bayesian updating replaces the implicit worst-case prior of frequentist hypergeometric tests with a transparent, data-driven belief distribution that evolves as screening progresses. This provides more interpretable and flexible stopping behavior than existing approaches.

3.5 Computational Implementation

To make inference computationally feasible, the posterior is evaluated on a discrete grid of possible values for K and ω. The full posterior matrix is updated at each screening step using vectorized matrix operations in R, which allows simultaneous evaluation across all parameter combinations and avoids the computational cost of nested loops (Gelman et al., 2013). For computational tractability, K is evaluated on a coarse grid with step size 20, yielding approximately 500 grid points for a corpus of $N = 10{,}000$, and ω is evaluated on a grid of 15 uniformly spaced points between 1 and 6. This discretization introduces a small approximation error relative to a full-resolution grid, but substantially reduces computational cost and enables large-scale Monte Carlo evaluation. The step size was chosen empirically to balance precision against runtime; sensitivity to grid resolution is acknowledged as a limitation in Section 5.5.

At each step, the likelihood vector over the grid is computed using Wallenius' noncentral hypergeometric probability mass function as implemented in the `BiasedUrn` package (Fog, 2008a). The posterior matrix is then renormalized and the stopping criterion evaluated. Without vectorization, the sequential updates required for live screening would become computationally infeasible for large scale datasets.

3.6 Monte Carlo Simulation Study

The performance of the stopping rule is evaluated using Monte Carlo simulations. For each experimental configuration, synthetic screening sequences are generated using the biased urn model, and the stopping rule is applied to each sequence. Performance is summarized across simulations using two metrics. Recall at stopping is defined as $R_{n_{\text{stop}}} = k_{n_{\text{stop}}}/K$, and work saved is defined as:

$$\text{WorkSaved} = 1 - \frac{n_{\text{stop}}}{N} \tag{7}$$

A stopping rule is considered safe if it achieves the recall target τ in all (or a specified high proportion of) simulations, and efficient if work saved is large (Repke et al., 2026). These two objectives are in tension: a more conservative rule is safer but saves less work, mirroring the core trade-off identified in the empirical literature (M. W. Callaghan & Müller-Hansen, 2024).

3.7 Experimental Design

The experimental evaluation proceeds in three stages, each designed to assess a different aspect of the stopping rule's performance.

Stage 1: Controlled Simulation under Constant Bias. The baseline behavior of the stopping rule is first evaluated under controlled conditions where the model assumptions are satisfied. Synthetic screening sequences are generated using the biased urn model with four parameter configurations that vary corpus size, prevalence, and ranking strength. The baseline configuration uses $N = 10{,}000$, a prevalence of 5% ($K = 500$), and $\omega = 3$. Sensitivity analyses additionally examine $\omega = 5$ to assess performance under stronger ranking, prevalence of 1% to simulate sparse corpora typical of large-scale evidence synthesis, and $N = 5{,}000$ to evaluate behavior on smaller datasets. Each configuration is repeated across 100 Monte Carlo runs, and the stopping rule is evaluated across nine combinations of $\tau \in \{0.90, 0.95, 0.99\}$ and $\gamma \in \{0.90, 0.95, 0.99\}$, yielding $4 \times 9 = 36$ experimental conditions in total. Performance is measured using recall at stopping and work saved as defined in Section 3.6.

Stage 2: Robustness under Changing Bias. The second stage examines the behavior of the stopping rule when the constant bias assumption is violated. All robustness experiments use $N = 10{,}000$, prevalence of 5%, $\tau = 0.95$, and $\gamma = 0.99$, with 100 Monte Carlo runs per scenario. Four bias scenarios are implemented:

1. **Constant bias**: $\omega = 3$ throughout screening, corresponding to the model assumption.
2. **Linear decay**: ω decreases linearly from 5 to $5 - \lambda \times 3$, where λ is a decay strength parameter, representing gradual deterioration of ranking quality.
3. **Sudden drop**: $\omega = 5$ for the first 30% of documents, dropping to $\omega = 2$ thereafter, simulating abrupt ranking deterioration.
4. **Remaining relevance**: $\omega = 1 + 4 \times (R_{\text{remaining}}/K)$, so that ranking strength declines as fewer relevant documents remain.

The linear decay scenario is then extended to a sensitivity analysis over decay strength $\lambda \in \{0.25, 0.50, 0.75, 1.00\}$, and both the unadjusted and conservatively adjusted versions of the stopping rule are evaluated at each level, allowing the robustness benefit of the adjustment to be quantified as a function of violation severity.

Stage 3: Evaluation on Real-World Screening Data. The third stage evaluates the stopping rule on real-world systematic review screening data provided by M. Callaghan (2026) via Zenodo. [1] The Bayesian stopping rule and the frequentist baseline, specifically the hypergeometric stopping criterion of M. W. Callaghan and Müller-Hansen (2020), implemented with `bias = 1` (i.e. assuming random sampling without ranking bias correction), are applied to each run, and stopping points, achieved recall, and work saved are computed for both methods. This choice reflects the most common practical implementation of the criterion, in which the bias parameter is not known in advance, and creates a direct methodological contrast: the Bayesian method infers ω from the observed screening sequence, while the baseline ignores ranking bias entirely (Repke et al., 2026). A comparison against a bias-aware frequentist baseline such as BUSCAR (M. W. Callaghan & Müller-Hansen, 2024) is left for future work. The dataset comprises screening outputs from ten completed systematic reviews, structured by review and run. For each of the ten reviews, 100 independent screening runs are available, yielding 1,000 run-level observations per confidence level and 3,000 in total across $\gamma \in \{0.90, 0.95, 0.99\}$. Experiments were parallelized across runs and executed on a high-performance computing cluster to enable evaluation at scale. The reviews vary substantially in corpus size ($N = 258$ to $N = 8{,}793$), prevalence (0.3% to 8.2%), and effective ranking quality, providing a varied evaluation dataset for

[1] Dataset available at: https://zenodo.org/records/18164103

assessing the generalization of the simulation findings to realistic screening conditions.

Due to the computational cost of the exact Wallenius likelihood computation at scale, the PIK cluster implementation uses an approximate inference procedure based on exponential tilting of the hypergeometric likelihood, with ω estimated from the observed enrichment ratio at each screening step rather than inferred jointly with K via a full prior. This approximation was adopted for practical reasons: the exact grid-based procedure used in the R simulation experiments evaluates a full $(K \times \omega)$ posterior matrix at every screening step, which becomes computationally prohibitive for large corpora such as Review 6 ($N = 8{,}793$) across 100 independent runs and three confidence levels. The approximation enables evaluation at scale on the high-performance computing cluster while preserving the core inferential logic of the stopping rule. The methodological difference between the two implementations is acknowledged as a limitation in Section 5.5.

4 Results

4.1 Controlled Simulation under Constant Bias

Results from the Stage 1 Monte Carlo simulation are summarized in Figures 1, 2, and 3, which show median recall at stopping, median work saved, and miss rate respectively across all four experimental configurations and nine τ/γ combinations.

4.1.1 Recall at Stopping

Across all configurations where the stopping rule triggered, median recall increases monotonically with both τ and γ, confirming that the method behaves as intended: higher recall targets and confidence levels produce more conservative stopping and therefore higher achieved recall (Figure 1).

Under the baseline configuration ($N = 10{,}000$, prevalence = 5%, $\omega = 3$), median recall ranges from 0.923 at $\tau = 0.90$, $\gamma = 0.90$ to 0.992 at $\tau = 0.99$, confirming that the posterior correctly concentrates around the target level.

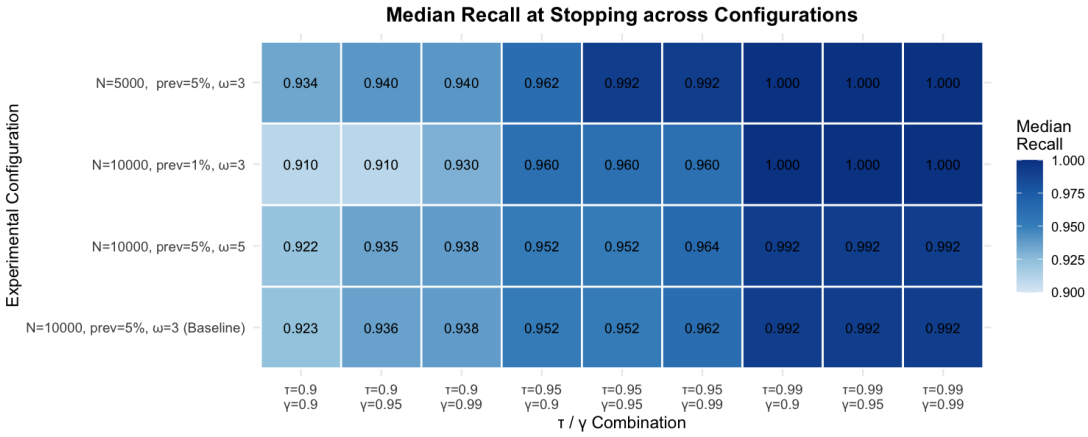

Figure 1: Median recall at stopping across four experimental configurations and nine combinations of recall target τ and confidence level γ. Each cell reports the median recall achieved across 100 Monte Carlo runs. Recall increases monotonically with both τ and γ across all configurations, confirming the method's intended behavior. The sparse prevalence configuration ($N = 10{,}000$, prevalence $= 1\%$) achieves a median recall of 1.000 at $\tau = 0.99$, reflecting rapid posterior concentration when the number of relevant documents is small.

4.1.2 Work Saved

Work saved decreases predictably as τ and γ increase, reflecting the fundamental trade-off between reliability and efficiency (Figure 2). Under the baseline configuration, median work saved ranges from 41.4% at $\tau = 0.90$, $\gamma = 0.90$ to 20.3% at $\tau = 0.99$. The stronger ranking configuration ($\omega = 5$) achieves substantially higher work savings across all parameter combinations, with median work saved of 57.7% at $\tau = 0.90$, $\gamma = 0.90$ and 37.6% at $\tau = 0.99$. This confirms that the method successfully exploits effective ML prioritization to stop earlier when ranking quality is high, consistent with the biased urn generative model (M. W. Callaghan & Müller-Hansen, 2024).

The smaller corpus configuration ($N = 5{,}000$) reveals an important boundary condition: at $\tau = 0.99$, the stopping rule fails to trigger in any of the 100 Monte Carlo runs, producing zero work saved. This indicates that a corpus of 5,000 documents with 5% prevalence is too small for the posterior to achieve the required 99% confidence in having met the recall target, given the resolution of the parameter grid. This finding highlights a practical limitation of the method: in small reviews with high recall requirements, the posterior may never become sufficiently concentrated to trigger stopping, and reviewers would need to screen the full corpus.

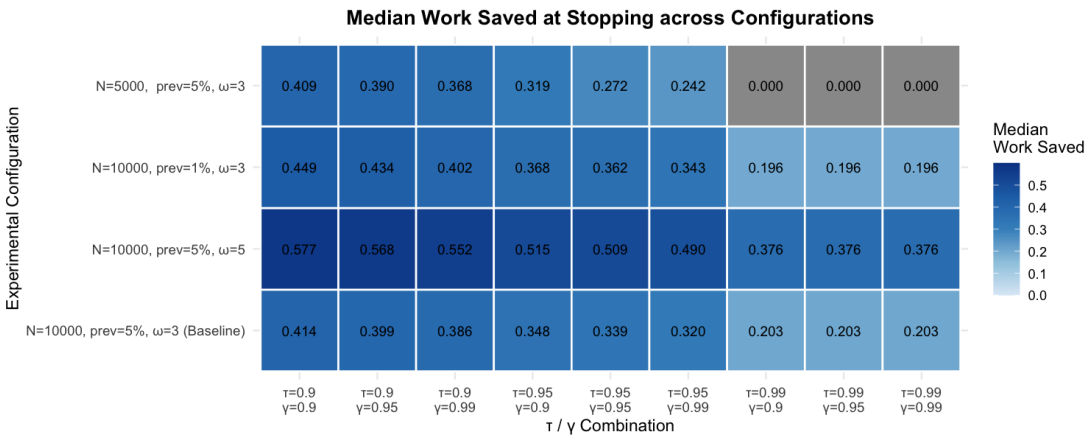

Figure 2: Median work saved at stopping across four experimental configurations and nine combinations of recall target τ and confidence level γ. Each cell reports the median proportion of documents not screened across 100 Monte Carlo runs. Work saved decreases as τ and γ increase, reflecting the fundamental trade-off between reliability and efficiency. The stronger ranking configuration ($\omega = 5$) achieves substantially higher work savings across all parameter combinations. Grey cells (value = 0.000) indicate configurations in which the stopping rule failed to trigger in any of the 100 runs, representing a boundary condition of the method.

4.1.3 Miss Rate and Guarantee Reliability

The miss rate heatmap (Figure 3) reveals that the reliability of the recall guarantee depends critically on the choice of γ. At $\gamma = 0.90$, miss rates are non-trivial across all configurations: under the baseline, 6% of runs miss the target at $\tau = 0.90$ and 8% miss it at $\tau = 0.95$.

These rates decrease substantially as γ increases, reaching zero for most configurations at $\gamma = 0.99$. The key practical implication is that $\gamma = 0.90$ provides insufficient protection against recall failures regardless of the recall target chosen, while $\gamma = 0.99$ produces reliable guarantees across all configurations where stopping triggers. The minimum practically reliable parameter combination appears to be $\tau = 0.95$, $\gamma = 0.99$, which achieves zero miss rate in all large-corpus configurations, though real-world work savings are substantially lower than simulation results suggest, as shown in Section 4.3.

The elevated miss rates observed across all τ/γ combinations for the $N = 5{,}000$ configuration suggest a more general difficulty beyond the complete stopping failure at $\tau = 0.99$. With only $K = 250$ relevant documents, the posterior over K evaluated on a grid with step size 20 lacks sufficient resolution to concentrate reliably around the true value, increasing the probability of premature stopping and consequent recall shortfalls. This coarse-grid effect is more pronounced in small corpora where the absolute number of relevant documents is low, and represents

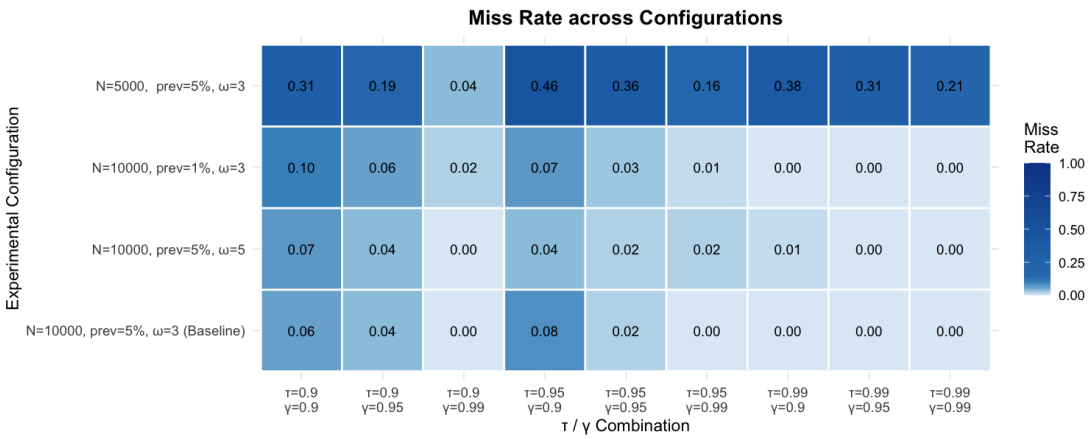

Figure 3: Miss rate across four experimental configurations and nine combinations of recall target τ and confidence level γ. Each cell reports the proportion of runs in which the stopping rule triggered but achieved recall below the target τ, across 100 Monte Carlo runs. The $N = 5{,}000$ configuration (top row) shows substantially elevated miss rates across all τ/γ combinations relative to the large-corpus configurations. This pattern reflects the difficulty of achieving posterior concentration in small corpora with limited relevant documents, as discussed in this section.

a practical limitation of the grid-based inference approach adopted here.

4.2 Simulation Study: Changing Bias

Results from the Stage 2 simulation are summarized in Table 1. Four bias scenarios were implemented as described in Section 3.7. Under the constant bias scenario, the method behaves as expected: median recall at stopping is 0.972 and the 1st percentile recall at stopping is 0.960, satisfying the target. Under linear decay, performance degrades modestly (1st percentile recall = 0.924) but the method continues to stop relatively early (median work saved = 0.453). The sudden drop and remaining relevance scenarios produce the largest violations of the recall target, with 1st percentile recall falling to 0.868 and 0.838 respectively. In these cases the inference model substantially overestimates ω late in screening, causing the posterior to become overly optimistic and triggering earlier stopping than warranted.

4.2.1 Sensitivity to Decay Strength

To investigate how strongly the constant bias assumption must be violated before the recall guarantee breaks down, the linear decay scenario was extended by

Table 1: Simulation results across bias scenarios ($\tau = 0.95$, $\gamma = 0.99$, $N = 10{,}000$, prevalence = 5%, $n = 100$ runs).

Scenario	Median Work Saved	Median Recall	Recall (1st Percentile)
Constant	0.286	0.972	0.960
Linear decay	0.453	0.954	0.924
Sudden drop	0.509	0.896	0.868
Remaining rel	0.398	0.876	0.838

varying the decay strength parameter $\lambda \in \{0.25, 0.50, 0.75, 1.00\}$. Under this parameterization, ω decreases linearly from 5 to $5-\lambda\times 3$ over the course of screening, so that $\lambda = 1$ corresponds to ω declining from 5 to 2 by the end of the review.

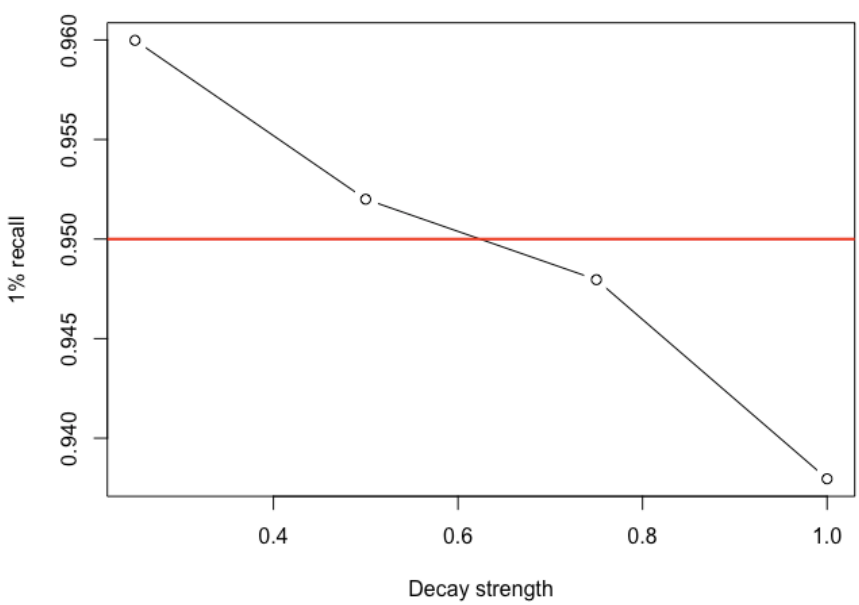

Figure 4: The 1st percentile recall at stopping as a function of decay strength λ under the linear decay scenario without conservative adjustment. The red horizontal line marks the $\tau = 0.95$ recall target.

Figure 4 shows the 1st percentile recall at stopping as a function of decay strength for the unadjusted method, with the red horizontal line marking the $\tau = 0.95$ recall target.

The results reveal a clear monotone relationship: as decay strength increases, the 1st percentile recall degrades. Notably, even at the mildest decay tested ($\lambda = 0.25$), the unadjusted method does not fully satisfy the guarantee, with the 1st percentile recall falling to 0.944. At maximum decay ($\lambda = 1.00$), the 1st percentile recall drops further to 0.924. This confirms that the stopping rule is

sensitive to even moderate violations of the constant bias assumption, and that the degree of degradation scales predictably with the severity of the violation.

4.2.2 Conservative Adjustment

To mitigate the sensitivity to bias misspecification, a conservative adjustment can be introduced in the Bayesian updating step by down-weighting the effective bias parameter by a factor $c \in (0, 1)$, so that $\tilde{\omega} = c \times \omega$ is used in the likelihood computation rather than ω directly. This causes the posterior to assign relatively more probability to lower values of ω, making the stopping criterion more conservative. A factor $c < 1$ thus provides a general mechanism for correcting systematic shifts between the assumed bias and the actual ranking process. Here, $c = 0.8$ is used as an illustrative example to demonstrate the effect of the adjustment.

The same sensitivity analysis was repeated with the adjustment applied. Figure 5 shows both curves — unadjusted and adjusted — on the same axes. Several features of the plot are noteworthy. First, the conservative adjustment consistently raises the 1st percentile recall across all decay strengths tested, shifting the blue curve upward by approximately 0.015–0.020 relative to the unadjusted black curve. Second, the two curves are approximately parallel, indicating that the adjustment provides a roughly constant improvement in recall guarantee regardless of the severity of the violation — it is a general correction rather than one effective only at a particular decay regime. Third, and most practically significant, the adjusted method satisfies the $\tau = 0.95$ guarantee up to approximately $\lambda = 0.50$, compared to no decay strength for the unadjusted method. Beyond $\lambda = 0.5$, even the adjusted method crosses below the recall target, with the 1st percentile recall falling to approximately 0.938 at $\lambda = 1.00$.

4.3 Performance on Real-World Screening Data

The Bayesian stopping rule and the frequentist baseline were evaluated on ten systematic reviews from the dataset of M. Callaghan (2026), comprising 100 independent screening runs per review and three confidence levels ($\gamma \in \{0.90, 0.95, 0.99\}$), for a total of 3,000 evaluated runs. The results are summarized by confidence levels in Table 2 and by individual reviews in Table 3.

4.3.1 Reliability

The Bayesian stopping rule achieves lower miss rates than the baseline at every confidence level (Figure 6). At $\gamma = 0.90$, the Bayesian method misses the recall target in 1.5% of runs, compared to 5.1% for the baseline. At $\gamma = 0.95$, miss rates are 0.9% and 2.3% respectively, and at $\gamma = 0.99$ they converge to 0.1% and 0.2%. The Bayesian method thus consistently provides a tighter probabilistic guarantee on recall across all confidence levels tested, confirming its intended behavior as a reliability-first stopping rule.

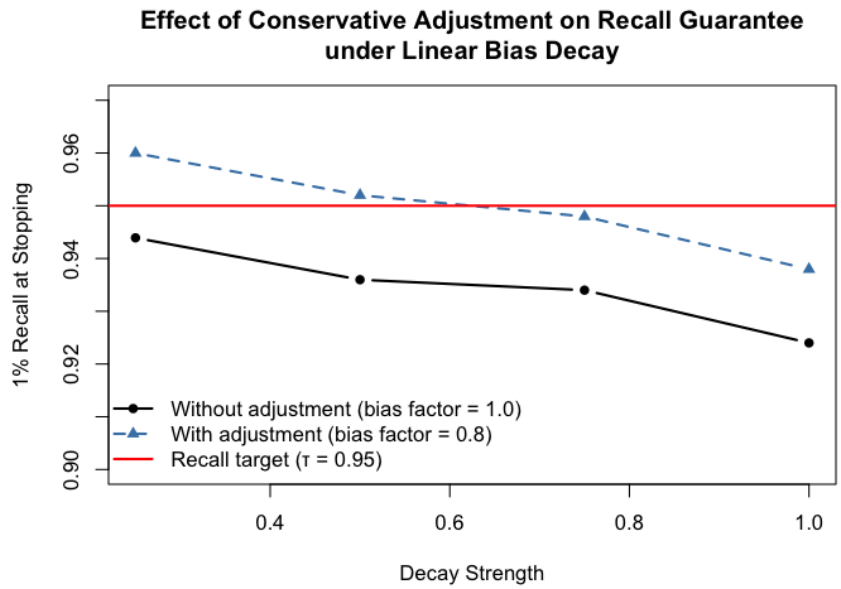

Figure 5: The 1st percentile recall at stopping as a function of decay strength λ for the unadjusted Bayesian method (solid black) and the conservatively adjusted method with bias factor 0.8 (dashed blue). The red horizontal line marks the $\tau = 0.95$ recall target. The conservative adjustment extends the range of decay strengths under which the recall guarantee is maintained.

4.3.2 Efficiency

The reliability advantage of the Bayesian method comes at a substantial efficiency cost (Figure 7). At $\gamma = 0.95$, the Bayesian method saves a mean of 1.5% of screening effort, compared to 17.6% for the baseline. This gap is consistent across all confidence levels and represents the primary trade-off of the proposed approach. Rather than a methodological shortcoming, this conservatism reflects the method's design: the Bayesian stopping rule maintains its guarantee regardless of ranking quality, while the baseline exploits high-quality rankings more aggressively.

4.3.3 Review-Level Heterogeneity

The aggregate efficiency gap masks substantial variation across individual reviews. Figure 8 shows the distribution of additional effort required by the Bayesian method relative to the baseline at $\gamma = 0.95$, revealing a markedly trimodal pattern. Table 3 provides the full breakdown.

Two reviews stand out as extreme cases. Review 6 (N = 8,793, prevalence = 1.2%) and Review 9 (N = 7,216, prevalence = 8.1%) see the baseline save approximately 51% and 50% of screening effort respectively, while the Bayesian

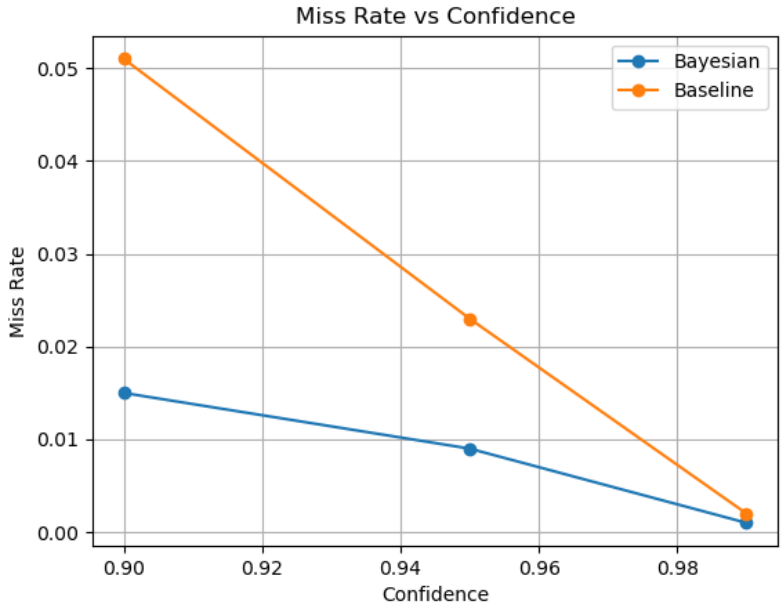

Figure 6: Miss rate as a function of confidence level (γ) for the Bayesian stopping rule and the frequentist baseline across all real-world reviews.

Table 2: Summary of miss rate and mean work saved by confidence level.

τ	Miss Rate (Bayesian)	Miss Rate (Baseline)	Work Saved (Bayesian)	Work Saved (Baseline)
0.90	0.015	0.051	0.019	0.208
0.95	0.009	0.023	0.015	0.176
0.99	0.001	0.002	0.010	0.120

method saves only 2.4% and 3.7%. These are also the two largest reviews in the dataset, and both exhibit strong ML prioritization as evidenced by the baseline's ability to stop very early. This pattern reveals a key dynamic: the better the ranking model, the more aggressively the baseline exploits it, while the Bayesian method remains conservative regardless of ranking performance. In the majority of reviews, however, where the baseline saves modest amounts of work (1–10%), the absolute efficiency gap between the two methods is small. The trimodal distribution in Figure 8 reflects precisely this split between reviews with very effective and moderately effective ML prioritization.

Figure 7: Mean work saved as a function of confidence level (γ) for the Bayesian stopping rule and the frequentist baseline.

5 Discussion

5.1 A Conservative but Trustworthy Stopping Criterion

The controlled simulation results under constant bias confirm that the method behaves as theoretically expected: recall at stopping is driven by τ and γ, work savings scale with ranking quality, and the guarantee is reliably satisfied at $\gamma = 0.99$ across all large-corpus configurations. These findings provide the interpretive foundation for the real-world evaluation: the efficiency gap between the Bayesian method and the baseline observed on the Callaghan dataset is consistent with the simulation results for moderate ranking strength ($\omega = 3$), while the reviews with very high work savings in the baseline (Reviews 6 and 9) correspond to the high-ranking scenario ($\omega = 5$) where the gap is largest.

The central finding is that the Bayesian stopping rule provides a more reliable recall guarantee than the frequentist baseline across all confidence levels and reviews tested, at the cost of substantially lower work savings. This trade-off is not incidental, as it reflects a deliberate design choice. The Bayesian method maintains its guarantee regardless of ranking quality, while the baseline, though it does not explicitly account for ML prioritization, tends to stop earlier in reviews with strong ranking because the observed sequence of predominantly non-relevant documents leads the hypergeometric model to estimate that few relevant documents remain (M. W. Callaghan & Müller-Hansen, 2020). This earlier stopping comes

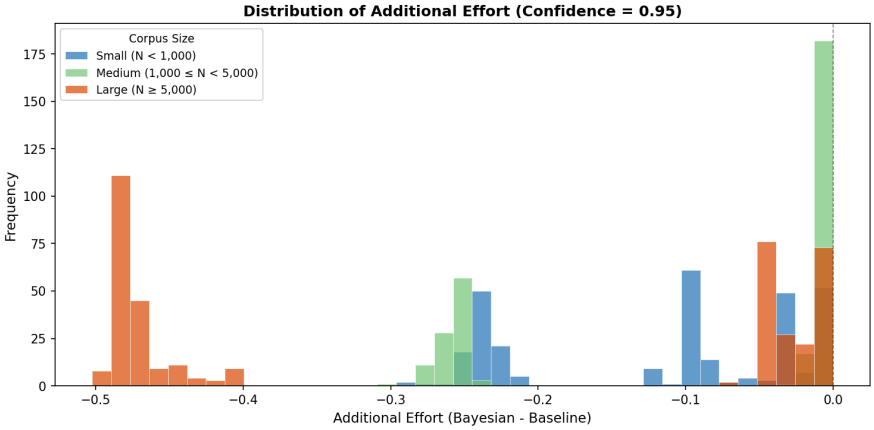

Figure 8: Distribution of additional screening effort required by the Bayesian method relative to the baseline ($\gamma = 0.95$). Negative values indicate that the Bayesian method screens more documents than the baseline. Bars are color coded by corpus size: small ($N < 1{,}000$, blue), medium ($1{,}000 \leq N < 5{,}000$, green), and large ($N \geq 5{,}000$, orange). The trimodal distribution reflects the concentration of reviews within each size category, with large reviews driving the left tail of high additional effort.

at the cost of weaker recall assurance, as the baseline does not account for the possibility that relevant documents may be concentrated earlier in the ranking (M. W. Callaghan & Müller-Hansen, 2020; Repke et al., 2026).

In absolute terms, the work savings achieved by the Bayesian method are modest at $\gamma = 0.95$, but in high-volume reviews even small percentage savings correspond to meaningful reductions in manual screening effort, while the method still provides a statistical guarantee that existing stopping rules cannot offer (M. W. Callaghan & Müller-Hansen, 2024; Repke et al., 2026). In contexts where missing a relevant study has serious downstream consequences — including systematic reviews informing clinical guidelines, legal document review, and hate speech moderation — the reliability advantage of the Bayesian approach may outweigh the efficiency cost (Cormack & Grossman, 2014; Stevenson & Bin-Hezam, 2023).

5.2 The Reliability–Efficiency Trade-off Across Reviews

The review-level analysis reveals that the efficiency gap between the two methods is not uniform but concentrated in reviews where the ML ranking is highly effective. Reviews 6 and 9, both large corpora with strong prioritization, account for the most extreme divergence: the baseline saves approximately half of all screening effort in these reviews, while the Bayesian method saves only 2–4%. The trimodal distribution of additional effort (Figure 8) clearly shows that for the majority of reviews, the baseline offers only modest efficiency gains, and the

Table 3: Per-review statistics at $\gamma = 0.95$, sorted by prevalence.

Review	N	Prevalence	Work Saved (Bayesian)	Work Saved (Baseline)
8	5,812	0.003	0.003	0.043
4	1,630	0.006	0.005	0.010
3	1,908	0.008	0.003	0.015
6	8,793	0.012	0.024	0.510
12	5,214	0.021	0.023	0.034
10	882	0.042	0.009	0.095
5	258	0.058	0.000	0.019
11	975	0.076	0.018	0.254
9	7,216	0.081	0.037	0.498
7	1,175	0.082	0.026	0.284

absolute difference between the two methods is small.

This pattern has an important practical implication: the contexts where practitioners most need a true statistical guarantee are precisely those where screening is difficult and ranking is less effective, meaning the efficiency cost of the Bayesian method will be correspondingly low in exactly those settings.

5.3 Robustness to Changing Bias and Conservative Adjustment

The simulation study reveals that the Bayesian stopping rule is sensitive to violations of its constant bias assumption, but in a structured and predictable way. Under gradual linear decay of ω, the guarantee degrades monotonically with the severity of the violation. Even at the mildest decay tested ($\lambda = 0.25$), the unadjusted method falls slightly below the recall target, with the 1st percentile recall at approximately 0.944. Strong decay ($\lambda \geq 0.75$) leads to more meaningful violations, with 1st percentile recall falling to approximately 0.924 at $\lambda = 1.00$.

These findings align with the concern raised in the literature that misspecification of the bias parameter can lead to overly optimistic stopping (M. W. Callaghan & Müller-Hansen, 2024). However, they also contextualize this concern: for moderate violations of the constant bias assumption — which are arguably more realistic than extreme sudden drops in ranking quality — the method remains approximately valid. The linear decay scenario with $\lambda \leq 0.25$ represents a situation where the ranking model gradually loses effectiveness as relevant documents become harder to find, consistent with the recall curve patterns observed by (Ferdinands et al., 2023).

Importantly, the sensitivity analysis serves a dual purpose. While it demonstrates that the method is sensitive to strong violations of the constant bias assumption, it also validates that assumption as a reasonable modeling choice. The alternative assumptions tested — gradual decay, sudden deterioration, and de-

pendence on remaining relevant documents — all produce worse outcomes than the constant bias model, both in terms of miss rate and stopping reliability. This suggests that assuming constant bias is not merely a convenient simplification, but a defensible and arguably conservative choice: by not attempting to model the dynamics of ω during screening, the inference procedure avoids overfitting to transient patterns in the ranking and maintains a more stable posterior. The advance over the frequentist status quo is therefore twofold: a less restrictive modeling assumption that produces a more conservative and trustworthy stopping criterion, at the cost of modest efficiency losses that are justified in contexts where the recall guarantee matters most.

To mitigate this sensitivity, the conservative adjustment described in Section 4.2.2 introduces a deliberate asymmetry into the inference procedure: by systematically underestimating ω, the posterior remains uncertain about recall for longer before triggering the stopping condition. This is conceptually analogous to the conservative prior embedded in frequentist hypergeometric stopping rules (M. W. Callaghan & Müller-Hansen, 2020), but operationalized within the Bayesian framework in a transparent and tunable way. With $c = 0.8$ as an illustrative example, the adjusted method satisfies the $\tau = 0.95$ guarantee up to approximately $\lambda = 0.50$, compared to no decay strength for the unadjusted method.

More generally, any factor $c \in (0, 1)$ can be used to introduce conservatism into the inference procedure, with smaller values producing more conservative stopping behavior at the cost of reduced efficiency. The choice of $c = 0.8$ is illustrative rather than prescriptive, and in practice the appropriate value will depend on the degree of model misspecification expected for a given review. Importantly, this correction factor is not a parameter that can be meaningfully inferred from the screening data itself, as it is not an unknown quantity about the data-generating process, but rather a deliberate adjustment to account for unknown deviations between the assumed model and the actual screening process. This distinguishes it from the bias parameter ω, which is a property of the ranking model and can in principle be estimated from observed screening outcomes. The correction factor c instead plays the role of a robustness tuning parameter, analogous to a regularization coefficient, whose value reflects the analyst's degree of distrust in the assumed bias model.

5.4 First Bayesian Stopping Rule for Systematic Review Screening

As established in Section 2.6, this thesis presents the first Bayesian stopping criterion for ML-assisted systematic review screening. Existing approaches — including the hypergeometric stopping rule (M. W. Callaghan & Müller-Hansen, 2020), BUSCAR (M. W. Callaghan & Müller-Hansen, 2024), and point process methods (Stevenson & Bin-Hezam, 2023) — are formulated as frequentist hy-

pothesis tests or statistical estimators. While Bron et al. (2025) and Stevenson and Bin-Hezam (2023) represent important recent advances in stopping methodology, neither adopts a Bayesian generative model that jointly infers both the total number of relevant documents and the ranking bias from the observed screening sequence.

The Bayesian framework offers three advantages over existing approaches that are demonstrated empirically in this thesis: lower miss rates at matched confidence levels; direct probabilistic statements about recall rather than hypothesis test outcomes; and a principled mechanism for introducing conservatism through prior or likelihood adjustment rather than through ad hoc parameter choices.

A further advantage of the Bayesian framework concerns the problem of multiple testing. Frequentist stopping rules based on hypothesis testing face a well-known theoretical concern: because the test is evaluated repeatedly at each screening step, the Type I error rate is not controlled at the nominal level if the rule is applied sequentially without correction (Berger, 1985). This problem is acknowledged by M. W. Callaghan and Müller-Hansen (2020) who note that the multiple testing issue was not formally addressed in their derivation and identify it as a direction for future work. This multiple testing problem arises because each evaluation of the stopping criterion constitutes a separate hypothesis test, and the probability of a false rejection accumulates across tests. The Bayesian stopping rule does not face this problem. Since it is grounded in posterior inference rather than repeated hypothesis testing, the stopping criterion at each step reflects the full accumulated evidence from all prior screening outcomes rather than a series of independent tests (Berger, 1985). The posterior probability $\Pr(R_n \geq \tau \mid y_{1:n})$ is a coherent summary of current beliefs about recall given all observed data, and its sequential updating does not inflate error rates in the way that repeated frequentist tests do (Berger, 1985; Gelman et al., 2013). This represents a theoretically important advantage of the Bayesian framework over existing frequentist stopping rules, particularly in the context of live systematic reviews where the stopping criterion may be evaluated hundreds or thousands of times before triggering.

5.5 Limitations and Future Work

Several limitations of the current implementation should be acknowledged. First, the conservative adjustment factor of 0.8 was chosen heuristically and evaluated at a single value. The choice of adjustment factor will in practice depend on the degree of model misspecification expected for a given review. Second, the simulation study uses synthetic data generated under the biased urn model, which may not capture the full complexity of real-world ranking behavior. The evaluation of the PIK cluster on ten real reviews provides empirical validation, but a larger and more diverse set of reviews would strengthen the generalizability of the findings. Third, the method fails to trigger stopping in small corpora ($N \approx 5{,}000$) at high recall targets ($\tau = 0.99$), suggesting that the coarse parameter grid used

for computational efficiency may be insufficient for the posterior to concentrate adequately in low-information settings. Future work should investigate adaptive grid refinement or importance sampling approaches to extend the method's applicability to smaller reviews (Gelman et al., 2013). Fourth, the current implementation assumes that the bias parameter ω is constant throughout screening and scalar. Extending the model to allow for time-varying or document-type-specific bias would address the sensitivity identified in the robustness analysis (M. W. Callaghan & Müller-Hansen, 2024; Repke et al., 2026). Fifth, future work might investigate measures to quantify how well real-world screening sequences are captured by the biased urn model, explore alternative priors or model modifications to improve efficiency in real-world settings, and develop computationally lighter implementations to enable deployment in live screening workflows.

A further limitation concerns the implementation used for the real-world evaluation on the PIK cluster. Due to computational constraints, the Python implementation used in Stage 3 approximates the Wallenius noncentral hypergeometric likelihood via exponential tilting rather than exact computation, and estimates ω from the observed enrichment ratio rather than placing a full prior over ω and updating it jointly with K (Fog, 2008a). This approximation diverges from the exact grid-based inference procedure used in the R simulation experiments, meaning the two sets of results are not directly comparable. The approximation error is likely largest for reviews with very strong ML ranking, which may contribute to the counterintuitive efficiency gap observed in Reviews 6 and 9. A unified implementation using exact Wallenius likelihood computation for both simulation and real-world evaluation is an important direction for future work.

Finally, the frequentist baseline used in the real-world evaluation assumes random sampling (`bias = 1`), corresponding to the original hypergeometric criterion of M. W. Callaghan and Müller-Hansen (2020). A stronger comparison would evaluate the Bayesian method against a bias-aware frequentist baseline such as BUSCAR (M. W. Callaghan & Müller-Hansen, 2024), which explicitly models ranking bias but requires the user to specify ω in advance. Such a comparison would more directly isolate the contribution of Bayesian inference over frequentist bias-aware stopping, and is a natural direction for future work.

Future work might also investigate the integration of the Bayesian stopping rule into existing screening platforms such as ASReview or NACSOS (Repke et al., 2026), where it could be offered alongside the baseline as a high-guarantee alternative for users who prioritize recall assurance over efficiency.

6 Conclusion

This thesis set out to address a longstanding methodological gap in ML-assisted systematic review screening: the absence of a stopping criterion grounded in a fully Bayesian framework. Building on the biased urn model introduced by

M. W. Callaghan and Müller-Hansen (M. W. Callaghan & Müller-Hansen, 2024), a Bayesian stopping rule was developed that jointly infers the total number of relevant documents and the ranking bias parameter from the observed screening sequence, and stops screening once the posterior probability of having achieved a target recall level exceeds a user-specified confidence threshold.

Three contributions were made. First, a vectorized computational implementation was developed enabling large-scale Bayesian inference over a joint parameter grid. Second, Monte Carlo simulations across four configurations demonstrated that the method reliably satisfies its recall guarantee at $\gamma = 0.99$ across all large-corpus settings. Third, real-world evaluation on ten systematic reviews confirmed consistently lower miss rates than the frequentist baseline at every confidence level tested.

The robustness analysis revealed that the method is sensitive to violations of its constant bias assumption, but in a structured and predictable way: mild violations preserve the guarantee, while strong violations lead to overly optimistic stopping. A conservative adjustment, implemented by down-weighting the effective bias parameter during inference, was shown to extend the range of assumption violations under which the guarantee is maintained, without requiring changes to the underlying model structure.

This thesis presents the first stopping criterion in this domain to adopt a Bayesian framework. The results demonstrate that such a framework offers genuine advantages over existing frequentist approaches in terms of reliability and interpretability, while the conservative adjustment provides a practically deployable mechanism for improving robustness under realistic conditions. The fundamental trade-off between recall assurance and efficiency identified here — and the finding that this trade-off is most pronounced precisely in the reviews where ML ranking is most effective — represents an important empirical contribution to the growing literature on stopping criteria for technology-assisted review (M. Callaghan et al., 2024; Repke et al., 2026).

References

Bastian, H., Glasziou, P., & Chalmers, I. (2010). Seventy-five trials and eleven systematic reviews a day: How will we ever keep up? *PLOS Medicine*, *7*(9), e1000326. https://doi.org/10.1371/journal.pmed.1000326

Berger, J. O. (1985). *Statistical decision theory and Bayesian analysis* (2nd). Springer.

Bron, M., van der Heijden, P. G. M., Feelders, A., & Siebes, A. (2025). Using Chao's estimator as a stopping criterion for technology-assisted review. *ACM Transactions on Information Systems*, *43*(3), 1–51. https://doi.org/10.1145/3724116

Callaghan, M. (2026). *Prioritised records from simulated prioritised screening on the Synergy dataset* [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.18164103

Callaghan, M., Müller-Hansen, F., Bond, M., Hamel, C., Devane, D., Kusa, W., O'Mara-Eves, A., Spijker, R., Stevenson, M., Stansfield, C., Thomas, J., & Minx, J. C. (2024). Computer-assisted screening in systematic evidence synthesis requires robust and well-evaluated stopping criteria. *Systematic Reviews*, *13*, 284. https://doi.org/10.1186/s13643-024-02699-7

Callaghan, M. W., & Müller-Hansen, F. (2020). Statistical stopping criteria for automated screening in systematic reviews. *Systematic Reviews*, *9*(1), 273. https://doi.org/10.1186/s13643-020-01521-4

Callaghan, M. W., & Müller-Hansen, F. (2024). *BUSCAR: Biased urns for efficient stopping criteria in technology-assisted reviews* [Manuscript. https://gitlab.pik-potsdam.de/mcc-apsis/biased-urn].

Cormack, G. V., & Grossman, M. R. (2014). Evaluation of machine-learning protocols for technology-assisted review in electronic discovery. *Proceedings of the 37th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 153–162. https://doi.org/10.1145/2600428.2609601

Ferdinands, G., Schram, R., de Bruin, J., Bagheri, A., Oberski, D. L., Tummers, L., Teijema, J. J., & van de Schoot, R. (2023). Performance of active learning models for screening prioritization in systematic reviews: A simulation study into the average time to discover relevant records. *Systematic Reviews*, *12*(1), 100. https://doi.org/10.1186/s13643-023-02257-7

Fog, A. (2008a). Calculation methods for Wallenius' noncentral hypergeometric distribution. *Communications in Statistics – Simulation and Computation*, *37*(2), 258–273. https://doi.org/10.1080/03610910701790269

Fog, A. (2008b). Sampling methods for Wallenius' and Fisher's noncentral hypergeometric distributions. *Communications in Statistics – Simulation and Computation*, *37*(2), 241–257. https://doi.org/10.1080/03610910701790236

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian data analysis* (3rd). Chapman; Hall/CRC.

Higgins, J. P. T., Thomas, J., Chandler, J., Cumpston, M., Li, T., Page, M. J., & Welch, V. A. (Eds.). (2019). *Cochrane handbook for systematic reviews of interventions* (2nd). John Wiley & Sons. https://doi.org/10.1002/9781119536604

Johnson, N. L., & Kotz, S. (1977). *Urn models and their application: An approach to modern discrete probability theory*. John Wiley & Sons.

O'Mara-Eves, A., Thomas, J., McNaught, J., Miwa, M., & Ananiadou, S. (2015). Using text mining for study identification in systematic reviews: A systematic review of current approaches. *Systematic Reviews*, *4*, 5. https://doi.org/10.1186/2046-4053-4-5

Repke, T., Tinsdeall, F., Danilenko, D., Graziosi, S., Müller-Hansen, F., Schmidt, L., Thomas, J., & van Valkenhoef, G. (2026). Don't stop me now, 'Cause I'm Having a Good Time Screening: Evaluation of stopping methods for safe use of priority screening in systematic reviews. *Cochrane Evidence Synthesis and Methods*, *4*(1), e70068. https://doi.org/10.1002/cesm.70068

Shemilt, I., Simon, A., Hollands, G. J., Marteau, T. M., Ogilvie, D., O'Mara-Eves, A., Kelly, M. P., & Thomas, J. (2013). Pinpointing needles in giant haystacks: Use of text mining to reduce impractical screening workload in extremely large scoping reviews. *Research Synthesis Methods*, *5*(1), 31–49. https://doi.org/10.1002/jrsm.1093

Stevenson, M., & Bin-Hezam, R. (2023). Stopping methods for technology-assisted reviews based on point processes. *ACM Transactions on Information Systems*, *42*(3), 1–37. https://doi.org/10.1145/3631990

Yang, E., MacAvaney, S., Lewis, D. D., & Frieder, O. (2022). Goldilocks: Just-right tuning of BERT for technology-assisted review. *Advances in Information Retrieval: 44th European Conference on IR Research (ECIR 2022)*, 502–517. https://doi.org/10.1007/978-3-030-99736-6_34

Appendix: Code Repository

All code, simulation scripts, data outputs, and figures associated with this thesis are publicly available in the following GitHub repository:

https://github.com/ashley-razo/Masters-Thesis.git

Scan the QR code or visit the URL above to access the repository.

Statement of Authorship

I hereby confirm and certify that this master thesis is my own work. All ideas and language of others are acknowledged in the text. All references and verbatim extracts are properly quoted and all other sources of information are specifically and clearly designated. I confirm that the digital copy of the master thesis that I submitted on April 27, 2026 is identical to the printed version I submitted to the Examination Office on April 27, 2026.

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Statement of Academic Integrity Related to the Use of Artificial Intelligence Tools

During the writing of this thesis, I made use of AI-based tools, including Claude (Anthropic), to support tasks such as code writing and debugging in R and Python, structuring and editing thesis sections, identifying areas of repetition or inconsistency, and refining the presentation of results and figures. The intellectual contributions of this thesis – including the research question, the Bayesian stopping rule framework, the design and execution of simulation experiments, the real-world evaluation on the PIK cluster, the interpretation of results, and all conclusions drawn – are entirely my own work.

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